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Extrafloral nectar production induced by simulated herbivory does not improve ant bodyguard attendance and ultimately plant defence

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Highly competitive and aggressive ant species are efficient bodyguards that monopolize the more attractive plants bearing extrafloral nectaries. Given that herbivory often increases the quality of extrafloral nectar, we hypothesized that plants damaged by herbivory would be more prone to interact with high-quality ant bodyguards and be better defended against herbivores. We performed an experiment with *Chamaecrista nictitans* plants. We induced anti-herbivore responses by applying jasmonic acid to a group of plants while keeping another group unmanaged. We measured extrafloral nectar production, censused ants visiting extrafloral nectaries and, subsequently, added herbivore mimics to measure the efficiency of ant anti-herbivore defence in both conditions. Induction increased the volume of extrafloral nectar and the mass of sugar per nectary without affecting the sugar concentration or the patterns of plant attendance and defence by ants. Thus, we found no evidence that defence-induced *C. nictitans* plants are more prone to interact with high-quality bodyguards or to receive better anti-herbivore defence. These findings highlight that increases in extrafloral nectar production are not always rewarded with increases in the biotic defences; instead, these rewards might be dependent on the traits of the nectar induced by herbivory events and/or on the ecological context in which the interaction is embedded. Consequently, herbivory might increase the costs of this induced biotic defence to plants bearing extrafloral nectaries when the induced defence does not increase the attractiveness of the plants to ants.

ADDITIONAL KEYWORDS: ant–plant interaction – *Chamaecrista nictitans* – defence mutualism – facultative mutualism – jasmonic acid.

INTRODUCTION

In facultative generalized mutualisms, each participant can interact with partners belonging to different species (Bronstein, 2015). These partner species differ in the net benefit provided to their mutualistic partners, with high-quality partners providing the highest net benefit among the other partners available within a community (West *et al.*, 1996; Bronstein, 2001). Thus, individuals interacting more often with high-quality partners should receive a higher net benefit than those interacting with low-quality partners (Stanton, 2003; Vázquez *et al.*, 2005). Considering that the occurrence and abundance of partners with different qualities vary within a community, it is expected that individuals engaged in generalized mutualisms will compete for high-quality partners (Jones *et al.*, 2012). In this scenario, individuals providing resources with higher relative value for partners or in larger quantities should be favoured because they are more attractive to high-quality mutualistic partners and to other mutualist partners in general (Cushman & Whitham, 1991; Fischer *et al.*, 2001; Palmer *et al.*, 2003). Therefore, the higher the attractiveness of the individual, the greater its probability of interacting with better mutualistic partners.

Protective mutualisms between plants bearing extrafloral nectaries (EFNs) and ants are examples of generalized mutualism in which both ants and

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plants differ in their quality as partners (Bronstein, 1998). In these mutualisms, plants have glands not directly related to pollination that produce nectar that attracts predators, mainly ants and occasionally other arthropods (Marazzi *et al.*, 2013; see Heil, 2015). Once on the plants, ants may repel or consume herbivores, reducing herbivory pressure on the plants (Bentley, 1977). Given that EFNs are easily accessible to many ant species, plants can be attended by several species that differ in the quality of anti-herbivore services (Xu & Chen, 2010; Nogueira *et al.*, 2015; Melati & Leal, 2018). More aggressive ant species recruit many workers and can locate and remove herbivores more efficiently from EFN-bearing plants than non-aggressive ant species that cannot recruit workers (Buckley & Gullan, 1991; Fiedler, 2001; Xu & Chen, 2010; Flores-Flores *et al.*, 2018; but see Melati & Leal, 2018). Moreover, more aggressive, numerically dominant ant species can easily monopolize resources in the habitat, influencing the behaviour of and driving away less aggressive ant species (Savolainen & Vepsäläinen, 1988). There is also a marked variation among plants in the quality and volume of extrafloral nectar offered to the ants. Within a community, plant species secreting nectar with higher relative values are generally more attractive to ants (see Koptur, 1994; Baker-Méio & Marquis, 2012; Nogueira *et al.*, 2015; Fagundes *et al.*, 2017). Given that the high-quality ant bodyguards for plants have a greater ability to monopolize plant resources, differences in EFN attractiveness might be an essential factor driving plant attendance patterns by high-quality bodyguards, potentially mediating the outcome of this interaction for plants.

One factor that can modify the attractiveness of EFN-bearing plants for ants is attack by herbivores on plant tissues. In several plant species, extrafloral nectar is an induced defence modulated by herbivory levels (optimal defence theory, McKey, 1974, 1979; also see Heil, 2015). Plants bearing EFNs have been reported to increase extrafloral nectar quality, quantity, or both after exposure to herbivory (Agrawal & Rutter, 1998; Wäckers *et al.*, 2001; Heil, 2015). This herbivory-induced investment in extrafloral nectar production can increase the attractiveness of the plants to ant bodyguards (Ness, 2003; Radhika *et al.*, 2008; Bixenmann *et al.*, 2011; Kwok & Laird, 2012). Therefore, plants attacked by herbivores could be more attractive to ant bodyguards than non-damaged neighbouring plants, reducing the risk of further herbivory (Heil *et al.*, 2000; Jones & Koptur, 2015; Hernandez-Cumplido *et al.*, 2016). Given that the most aggressive ants typically monopolize the highest-quality resources available, it is expected that the most aggressive, high-quality ant species will tend plants more often after herbivory events. If so, this could be one of the mechanisms favouring herbivory-induced extrafloral nectar secretion by EFN-bearing plants over evolutionary time, because the plants would receive higher anti-herbivore defence when facing herbivory.

In our study, we investigated the effect of induction of defences on the pattern of EFN-bearing plant attendance by high-quality ant bodyguards and its consequences for the quality of anti-herbivore services received by these plants. We hypothesized that herbivory events would increase plant attractiveness to high-quality ant bodyguards available in the habitat, leading to an increase in the quality of biotic defence that these plants received when under attack by herbivores. We used the plant *Chamaecrista nictitans* (Fabaceae), which bears large EFNs on its leaves, as a model to evaluate this hypothesis. We induced defensive responses by the plants by exogenous application of jasmonic acid on their leaves. Jasmonic acid is a volatile plant hormone responsible for inducing anti-herbivore defences, including the induction of extrafloral nectar that can increase biotic defences (Heil *et al.*, 2001; Heil, 2004). Most evidence about induction of extrafloral nectar by herbivore attacks comes from studies involving a few plant species from a reduced phylogenetic background (Agrawal & Rutter, 1998; Heil, 2015; references within). Therefore, induction of extrafloral nectar secretion by herbivory is not necessarily a widespread phenomenon among all plant lineages. For this reason, before evaluating our hypothesis, we assessed the assumption that individuals of *C. nictitans* increase their investment in extrafloral nectar production in response to exogenous jasmonic acid application.

MATERIAL AND METHODS

Model plant species

Chamaecrista nictitans (Fabaceae) is an annual, self-compatible, small woody shrub (~40 cm in height), with flowers bearing poricidal anthers pollinated by bees capable of floral sonication (Gottsberger & Silberbauer-Gottsberger, 1988; Lee, 1989). *Chamaecrista nictitans* plants are commonly found in ruderal areas and are widely distributed in the Americas, including Brazil (Gargiullo & Magnisson, 2008). Individuals of this species have EFNs on the adaxial surface of the leaf petioles (Gleason & Cronquist, 1991). Production of extrafloral nectar begins during leaf expansion and lasts until leaf senescence (A. Nogueira, personal observation). Although the extrafloral nectar of *C. nictitans* is consumed mostly by ant species, spiders, wasps and bees have also been observed feeding on their EFNs (Ruhren & Handel, 1999; Ruhren, 2003).

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Plant source and sampling design

To evaluate our hypothesis and assumption, we used plants cultivated from seeds in the greenhouse within the campus of Universidade Federal do ABC (UFABC), São Paulo, Brazil (23°40′S, 46°33′W), in April 2019. The region where the campus is located has an average annual temperature of 20.3°C and annual precipitation of 1340 mm. The seeds were harvested from mature plants belonging to a source population cultivated by our study group in the same area where our study was performed. The plants from this source population had not been submitted to any previous experimental treatments and had been cultivated only as a source of seeds. The source population was composed of individuals transplanted from two localities: (1) Parque Estadual do Juquery, a protected area of Cerrado vegetation in São Paulo state, Brazil (23°19′S, 46°41′W); and (2) a disturbed area occupied mainly by ruderal vegetation within the Federal University of ABC, São Paulo state, Brazil (23°40′S, 46°33′W). The second locality was the same area where we carried out this study (study site, hereafter). Individuals of *C. nictitans* maintained in the source population were transplanted from their natural occurrence area to pots in December 2017, during the peak of the rainy season. After they reached the reproductive stage, we collected mature fruits for this study. We carefully separated intact seeds from ten maternal plants from the source population.

To cultivate the plants used in this study, we first scarified the seeds from the ten maternal plants using sandpaper to break physical seed dormancy. After scarification, we planted three seeds in 8 L pots filled with compost soil formed by a mixture of 70% sand and 30% garden soil rich in organic matter and macronutrients (Terra Vegetal ABC GARDEN 20 kg). The same compost soil was used previously to cultivate the source population. Seeds from the same maternal plants were used to produce plants in three distinct pots, totalling 30 pots grown from ten different maternal plants. Five weeks after sowing, we removed seedlings with a smaller stature and less expanded leaves from the pots in which more than one seed had germinated, keeping only one seedling per pot. In the three pots in which no seedlings grew, we transplanted the spare seedlings from the same maternal plant germinated in other pots. During the first 60 days of growth of the plants, individuals were kept in natural light conditions and irrigated twice a day. By the end of this period, all plants had at least ten expanded leaves bearing EFNs secreting nectar.

AQ5

After 60 days of cultivation, we divided the 30 juvenile plants into two sets of plants. The first set of plants (N = 10 plants) was kept in the greenhouse to test our assumption that extrafloral nectar production is an inducible defence. All the plants of this first set came from five maternal plants. The second set of plants (N = 20 plants) was transferred to the field area surrounding the greenhouse in which the plants were cultivated. The plants in this second set came from ten maternal plants, five of which were the same maternal plants used for the first set of plants. In the experiments both inside and outside the greenhouse, we subdivided the plants into two treatments: (1) 15 plants (five in the first set and ten in the second) in which we induced anti-herbivore defences (induced defences, hereafter) with the application of jasmonic acid (JA treatment); and (2) 15 (five in the first set and ten in the second) unmanipulated plants (control). We raffled plants for each treatment, preventing individuals of the same maternal origin from being allocated in the same treatment.

For the JA treatment, we used a 0.1 mmol/L solution of JA prepared by solubilizing solid JA in 5% ethanol and diluting the concentrated alcoholic JA preparation with distilled water. Exogenous JA application has been used previously to elicit plant-induced defences against herbivores, including biotic defences (Baldwin *et al.*, 1997; Ament *et al.*, 2004; Cipollini *et al.*, 2004). In the case of extrafloral nectar secretion, it has been reported that the exogenous application of JA aqueous solutions to the plants elicits a similar response in EFN-bearing plants to true herbivory (Agrawal & Rutter, 1998; Heil, 2015; references therein). To investigate the potential phytotoxic impact of the JA solution in our plants, we performed a pilot study, in which we applied the JA solution onto the leaves of spare plants. We did not observe any phytotoxic effects for 10 days consecutively after the JA application. In all control plants, we applied an alcoholic solution of 5% ethanol diluted in distilled water (control solution hereafter). For both treatments, we applied the solutions onto the plants as a whole, spraying all leaves.

Evaluation of the induced response of extrafloral nectar production

The first set of plants (five plants in each group), used to evaluate patterns of extrafloral nectar secretion, were maintained in the greenhouse and irrigated twice a day. Owing to the high variation in production of extrafloral nectar among plants (e.g. Nogueira *et al.*, 2015), which could hinder our capacity to detect the effects of JA (see Heil *et al.*, 2000), we sampled the extrafloral nectar production twice in the same plant: before and after (1 day later) the application of the treatments on the plants of each group.

Before the first nectar sampling, we sprayed all ten plants with the control solution. Before applying it, we washed the EFNs on the leaves with distilled water to remove accumulated extrafloral nectar that could modify the nectar secretion afterwards. We bagged

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the whole plants 30 min after applying the control solution and kept them bagged for 12 h, from 18:00 to 06:00 h. We kept the plants bagged throughout the night period to prevent any undesirable effect of the bags on the photosynthesis of the plants and, consequently, on the extrafloral nectar secretion. After this period, we removed the bags and collected the nectar accumulated in each active foliar EFN using 1 µL microcaps (BRAND disposable BLAUBRAND).

The second nectar sampling occurred 24 h after the first sampling. Plants undergoing the JA treatment were sprayed twice with the JA solution on all leaves, with a 30 min interval between applications (Radhika *et al.*, 2008, 2010). Control plants were sprayed twice with the control solution. with a gap of 30 min between applications. While applying the solutions, we kept plants from the JA group momentarily in a separate room of the greenhouse to avoid any interference of this volatile hormone on the patterns of extrafloral nectar secretion by the control plants. Plants from the JA group were transferred back to the same room as the control plants after being bagged, because the volatile substances were retained within the bags. We also kept plants from the JA group apart from each other by ≥ 50 cm, to avoid plants from being affected by the JA volatiles released by neighbouring plants (Heil & Adame-Álvarez, 2010). We followed the same procedures used in the first nectar sampling, washing the EFNs before applying the solutions, bagging the plants for 12 h, and sampling the nectar using 1 µL microcaps. The first and the second nectar samplings were conducted in similar climatic conditions: 27.3 ± 3.1°C (first sampling) and 27.2 ± 2.6°C (second sampling), (mean ± SE); and 68% (first sampling) and 67% (second sampling) humidity at the beginning of the samplings.

In all nectar samples, we estimated the volume of nectar by measuring the proportion of each 1 µL microcap occupied by the nectar secreted from each active EFN with callipers (in millimetres) and converting it to microlitres. We estimated the average volume of nectar produced by each nectary by dividing the total volume of nectar collected from the whole plant by the number of active nectaries in each plant (i.e. EFNs producing extrafloral nectar in the plant). We measured the concentration of extrafloral nectar secreted (in degrees Brix) using a portable low-volume Eclipse refractometer (Bellingham & Stanley, Royal Tunbridge Wells, UK). For that, we used a minimal sample of 0.15 µL of extrafloral nectar collected from each plant, which was the minimal volume for the detection of concentration by the refractometer (H.D. Chinarelli, personal observation). We extracted the minimal nectar volume from one EFN located on the most apical branch in 11 plants, considering both treatments. In the other nine plants, the EFN located on the most apical branch did not secrete enough nectar, and we pooled the extrafloral nectar secreted by the subsequent EFNs until we reached ≥ 0.15 µL. In one plant, the volume of nectar secreted by all EFNs was insufficient for us to measure the sugar concentration. In this case, we added 1 µL of distilled water to the tiny nectar sample, measured the diluted nectar concentration in the refractometer and calculated its initial concentration using the dilution equation. We also calculated the mass of sugar (in milligrams) in the nectar samples as a proxy of the sugar content of the nectar. To do so, we first converted the nectar concentration from degrees Brix to milligrams per microlitre, according to Dafni (1992). Then, we used the sugar concentration in milligrams per microlitre of nectar and multiplied it by the volume of extrafloral nectar produced per EFN to estimate the average mass of sugar secreted per EFN in each plant.

Identification of high-quality ant bodyguards based on ant aggressiveness

To classify the quality of the local ant species as bodyguards of *C. nictitans*, we evaluated ant aggressiveness in the field around the greenhouse where we established our experimental set-up. Aggressiveness is an ant trait directly related to the competitive ability of a species within a community (Cerdá *et al.*, 2013), and the aggressiveness of different ant species towards other competitive ant species predicts the behaviour of these ants towards herbivores on plants (Fiedler, 2001; Ness, 2006; Xu & Chen, 2010).

We evaluated ant species aggressiveness by observing the behaviour of workers of different ant species when simultaneously exploring a standardized amount of resource (mixed baits). For that, we placed 30 mixed baits (canned sardine and honey) over 10 cm × 10 cm paper squares positioned on the ground in our study site, following Parr & Gibb (2010) and Melati & Leal (2018). We inspected the baits sequentially for 4 h (14:00–18:00 h) when the ants were more active at our study site (H. D. Chinarelli, personal observation). We observed the bait continuously for 3 min at each inspection and counted the number of ant workers of each morphospecies feeding on the bait. When there was more than one morphospecies on the bait, we extended the inspection to 5 min and counted the number of aggressive behaviours between workers of the different morphospecies. We considered that an aggressive ant behaviour occurred any time an ant bit, shoved, stung or expelled formic acid on other ants (Melati & Leal, 2018). At the end of each bait inspection, we collected one ant worker of each morphospecies for taxonomic identification.

We estimated the aggressiveness index for each ant morphospecies occurring in our study area by dividing

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the number of aggressive behaviours displayed by workers of one morphospecies by the number of baits in which this morphospecies interacted with other ant species (see Melati & Leal, 2018). The higher the ant aggressiveness index, the higher the probability of the workers belonging to this morphospecies behaving aggressively toward other ants and other arthropods on the plants, including herbivores. Consequently, we considered the ant morphospecies that scored the highest aggressiveness index value as the high-quality ant bodyguard at our study site.

Evaluation of the effect of induction of nectar secretion on plant attractiveness and efficiency of anti-herbivore defence

To evaluate how induction of extrafloral nectar secretion affects plant attractiveness to high-quality ant bodyguards and influences the quality of the anti-herbivore services received by the plants, the 20 plants in the field around the greenhouse were placed systematically ≥ 4 m apart from each other. After the plants had been in the field for 1 week, we sprayed the JA solution on ten plants (JA treatment) and the control solution on the other ten plants (control plants), following the same method described in the subsection 'Evaluation of the induced response of extrafloral nectar production'. After applying the JA or control solution, we counted the number of ants of each morphospecies foraging on each plant during 1 min, twice a day, for 5 days consecutively. The first census of ants started 1 h after application of the JA or control solution in the morning (10:00–12:00 h) and was repeated in the afternoon (14:00–18:00 h). For ant identification, we collected one worker of each morphospecies only in the census carried out in the morning. Given that all ants observed on the plants in the afternoon were also attending the plants in the morning, we did not collect ants in the afternoon, in order to minimize interference with ant behaviour in our subsequent ant efficiency tests on plant defence (described below).

After each observation of ant attendance in the afternoon (14:00–18:00 h), we evaluated the efficiency with which ants were defending plants in each treatment. We observed the behaviour of attending ants toward herbivore mimics in plants in the control and JA treatment groups. We used larvae of the beetle *Ulomoides dermestoides* (Coleoptera: Tenebrionidae) as a surrogate for herbivore presence on *C. nictitans*. These larvae are easily manipulated in the laboratory and on the plants, where they can move freely on the leaves and branches (for more information, see Passos & Leal, 2019). We placed one larva on the youngest leaf closest to the main branch apex of each plant, where EFNs tend to be more attractive to ants (Nogueira *et al.*, 2020). We performed this evaluation in the afternoon for logistic reasons. We also did not observe practical differences in ant attendance patterns occurring in the morning and the afternoon during pilot tests that we performed before the experiments (H. D. Chinarelli, personal observation). The efficiency of ant defence against the herbivore mimic was assessed once per day for each plant for 5 days, with a total of five larvae added per plant (one per day). Considering the 20 plants used in this experiment (ten control and ten JA-treated plants), we used a total of 100 larvae across plants and days.

After placement of the larva, we observed the plant until an ant morphospecies encountered the larva or for ≤ 10 min. If the larva was not encountered within 10 min, we considered that ants did not encounter the larva. When ants encountered the larva, we registered the time elapsed for the ant to encounter the larva and continued to observe ant behaviour for an additional 10 min. We classified the defensive role of ant morphospecies on plants as the frequency of the following: (1) failure, whereby the ant encountered the larva but did not exhibit any aggressive behaviour, or the ant exhibited aggressive behaviours (bit, shoved, stung or expelled formic acid) toward the larva but did not remove the larva from the plant; or (2) success, when the ant successfully killed or removed the larva from the plant. When the larva was removed from the plant, we registered the time elapsed from the first encounter of the larva with ants until its removal. After each observation, when ants encountered larvae we discarded the larvae to avoid any possible influence that the contact with ants could have on the next observations.

Data analysis

Effect of induction on the production of extrafloral nectar

To evaluate whether the application of JA influenced extrafloral nectar production, we built linear mixed models (LMMs) using induction (JA treatment and control), sampling time (first and second samplings) and the interaction between them as categorical factors. We used the average volume of nectar per EFN (in microlitres), the nectar concentration (in degrees Brix), and the average sugar mass (in milligrams) per EFN as response variables in the models. We considered adding temperature as a random variable, given its possible influence on extrafloral nectar volume and sugar concentration. However, we included only plant and maternal plant identities as random variables because none of the extrafloral nectar measurements was correlated with temperature (Supporting Information, Table S1;

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Fig. S1). We made planned orthogonal comparisons whenever the saturated model differed from the null models. In these cases, we compared the extrafloral nectar descriptors between plants that underwent herbivory simulation and unmanipulated plants [JA treatment in the second sampling (N = 5) > JA treatment in the first sampling + control plants in first and second samplings (N = 15)]. We performed these comparisons because the plants allocated to JA treatment had not yet received JA application in the first sampling, meaning that they were also untreated at the beginning of the experiment.

Effect of induction on patterns of ant attendance and efficiency as bodyguards on the plants

To evaluate whether JA application affected the probability of plant attendance by any ant species and the total number of ants attending these plants, we built generalized linear mixed models (GLMMs) with binomial and Poisson distributions, respectively. In the binomial model, we used the occurrence of ants on the plants (1 = presence; 0 = absence of ants) as the response variable and induction (JA treatment and control) as the categorical predictor factor. The Poisson model used the number of workers observed on the plants as the response variable and the induction (JA treatment and control) as the categorical predictor factor. Given that we were interested in comparing the recruitment of workers between the treatment groups, we did not consider in this last analysis the observations in which no ants attended the plants (N = 59 censuses). We also performed a separate GLMM with Poisson distribution to evaluate whether the induction had a disproportional influence on the number of workers on the plants belonging to different morphospecies. In this model, we used the number of ants of each morphospecies as the response variable and the induction, ant morphospecies and the interaction between these variables as the predictor variables. For this model, we also did not consider censuses with no ants from the morphospecies analysed. We also included plant identity, maternal plant identity, day and period (morning and afternoon) of the censuses as random variables in all models described above.

To evaluate whether JA application influenced the occurrence of high-quality ant bodyguards on *C. nictitans* plants, we performed GLMMs using binomial distributions and induction (JA treatment and control) as the categorical predictor factor. In this case, we used the occurrence of the most aggressive ant morphospecies (*Nylanderia* sp.; see Results) on the plants as the response variable (1 = presence of *Nylanderia* sp. ants; 0 = absence of *Nylanderia* sp. ants). For this model, we considered only those observations in which there was at least one worker from any ant species on the plant. We included plant identity, maternal plant identity, day and period (morning and afternoon) as random variables for all models above. The temperature was not considered in any models because ant attendance was not biased by its variation (for more detailed information, see Supporting Information, Figs S2 and S3).

Finally, we also performed a permutational multivariate analysis of variance (PERMANOVA) to investigate whether the induction of extrafloral nectar secretion affected the composition of ant species visiting the plants. To do so, we used a dissimilarity matrix based on the Bray–Curtis distance for ant species composition visiting the plants of the JA treatment and control groups. We built the dissimilarity matrix using the occurrence data of different ant morphospecies on the plants of each group throughout the ten observations. We used the group (JA treatment and control) as the grouping factor. To represent the composition of ant species visiting the plants graphically, we performed a non-metric multidimensional scaling (NMDS) using the aforementioned matrix of dissimilarity distance and grouping factor.

To investigate the effects of JA application on the quality of anti-herbivore service received by the plants, we built four models (two GLMMs and two LMMs), including the occurrence of induction (JA treatment and control) as the categorical predictor factor. In the first GLMM, we used a binomial distribution and included location of the larva (herbivore mimic) by ants as the response variable (1 = encountered; 0 = not encountered). In the second GLMM, we used a binomial distribution and included removal of the larva by ants as a response variable (1 = removed; 0 = not removed). In the third model (LMM), we used a Gaussian random distribution and the time elapsed before the ants encountered the larva as the response variable. In the fourth model (LMM), we also used a Gaussian distribution and included the time elapsed between encounter of the larva and its removal from the plant as a response variable. We included plant identity, maternal plant identity and day as random variables in all these four models. We did not include observations in which ants did not encounter the larva for the models predicting removal of the larva from the plants (second and fourth models).

Calculation of probabilities and software

We calculated probability values for all models using the likelihood ratio test, contrasting the model containing all variables (saturated model) against the null model with no fixed predictor factors (Zurr *et al.*, 2009). We performed all model analyses using the package *lme4* (Bates *et al.*, 2015) for the software R (R Core Team, 2017).

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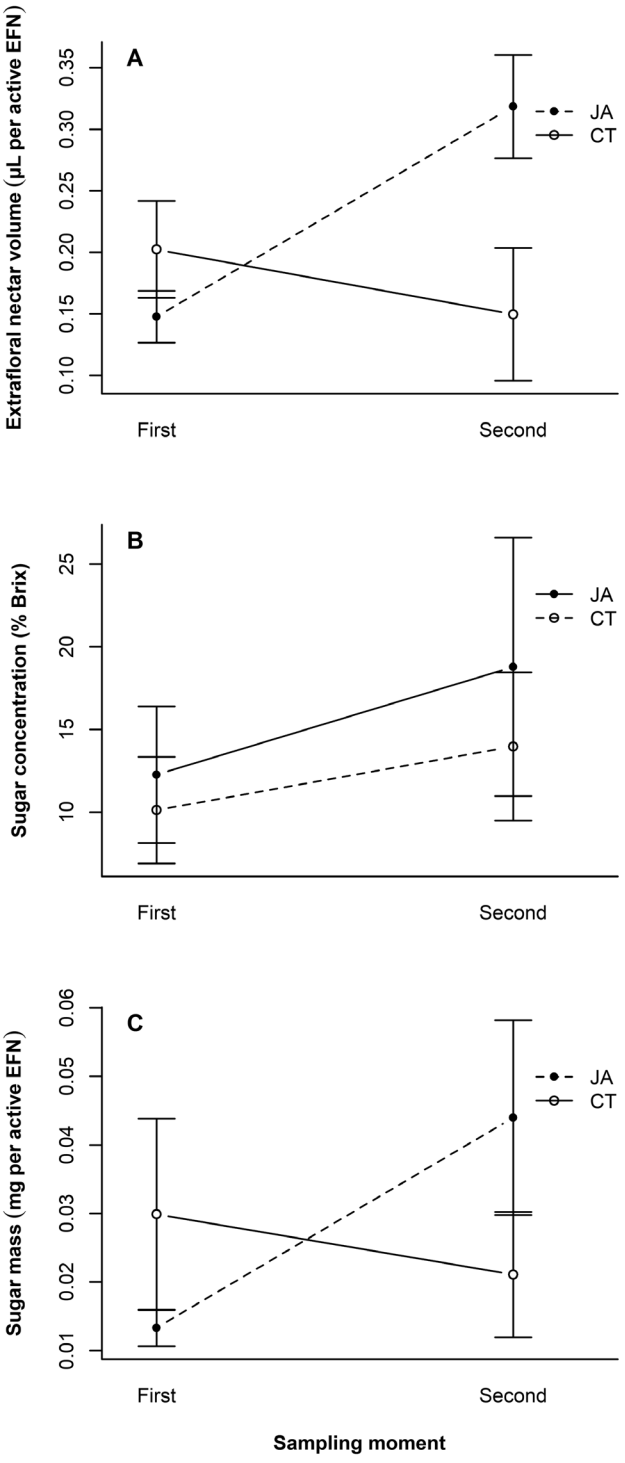


Figure 1. Influence of simulated herbivory on the characteristics of extrafloral nectar secreted by *Chamaecrista nictitans* (Fabaceae) (N = 20). A, volume of extrafloral nectar per active extrafloral nectary (EFN; in microlitres per active EFN; saturated model × moment + treatment model: χ^2 = 9.43; d.f. = 1; P = 0.0021; planned comparisons: z = 2.28; P = 0.023). B, extrafloral nectar sugar concentration (in degrees Brix; saturated model × null model; χ^2 = 4.70; d.f. = 3; P = 0.19). C, total sugar (in milligrams per active EFN; saturated model × moment + treatment model: χ^2 = 7.99; d.f. = 1; P = 0.0047; planned comparisons: z = 6.31; P < 0.001). In the first sampling, both plant groups had the extrafloral nectar sampled before the simulation of herbivory. In the second sampling, we simulated herbivory in half of the plants by exogenous application of jasmonic acid (JA treatment), while the other plants were treated as the control group (CT). Circles represent the mean value and whiskers the Standard Error (SE).

Predictions

If *C. nictitans* plants produce more extrafloral nectar after herbivory events (assumption), we expect that plants undergoing JA treatment will secrete a larger volume of more concentrated nectar, containing a larger sugar mass in comparison to unmanipulated control plants. If *C. nictitans* plants become more attractive to the best partner available after herbivory events, we expect that plants undergoing JA treatment will have a higher probability of being attended by the most aggressive ant species available in the study area and/or attended by more workers of these species. Finally, if *C. nictitans* plants increase the efficiency with which ants defend plants after herbivory events, we expect that plants undergoing JA treatment will have a higher probability of ants encountering a larva (herbivore mimic) and removing it. We also expect that both encounter and removal of a larva will be faster in JA plants than in control plants.

RESULTS

Induced response of extrafloral nectar production

All plants of *C. nictitans* produced extrafloral nectar throughout the greenhouse experiment. There was an increase in the volume of extrafloral nectar and mass of sugar for plants allocated to JA treatment after application of the JA solution (second sampling; Fig. 1A, C). However, there were no changes in the concentration of extrafloral nectar of JA plants in the second sampling compared with control plants (Fig. 1B). Regarding the differences between the extrafloral nectar secreted by JA and control plants in the second sampling, the nectar volume per active EFN was 91% higher for plants in the JA treatment than for the control plants (data $\log_{10}$-transformed to improve model fit; N = 10; saturated model × moment + treatment model: χ^2 = 9.43; d.f. = 1; P = 0.0021; Supporting Information, Table S2; Planned comparisons: z = 2.28; P = 0.023; Fig. 1A). The sugar concentration was similar between

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plants allocated to JA treatment and control plants, 13.96 ± 0.26 (mean ± SD) and 13.72 ± 11.87°Brix, respectively (N = 10; saturated model × null model: χ^2 = 4.70; d.f. = 3; P = 0.19; Supporting Information, Tables S2 and S3; Fig. 1B). Finally, the sugar mass of the extrafloral nectar per EFN was 105% higher in JA plants in the second sampling than in control plants ($\log_{10}$-transformed data to improve model fit; N = 10; saturated model × model without interaction between treatment and moment: χ^2 = 7.99; d.f. = 1; P = 0.0047; Table 1; Supporting Information, Table S2; and planned comparisons: z = 6.31; P < 0.001; Fig. 1C).

Ant aggressiveness index and quality of local ant morphospecies as bodyguards

We observed ant workers attending all 30 baits placed on the ground in the field. In 42 censuses (70%), workers of more than one ant morphospecies were feeding on the bait simultaneously. We observed workers of all four morphospecies that visited the plants (*Camponotus* sp., *Linepithema* sp., *Brachymyrmex* sp. and *Nylanderia* sp.) feeding on the baits (Table 2). In contrast, workers of four other ant morphospecies fed exclusively on the baits and were not observed attending the plants (*Dorymyrmex* sp., *Pheidole* sp., *Solenopsis* sp. and *Wasmannia* sp.). Among the ants attending the plants, *Linepithema* sp. exhibited the lowest aggressiveness index (0.08), followed by *Camponotus* sp. (0.33), *Brachymyrmex* sp. (0.65) and *Nylanderia* sp. (0.82). *Nylanderia* sp. was not only the most aggressive ant attending our plants, but also the most aggressive ant species exploring the baits (Table 2). For this reason, we classified *Nylanderia* sp. as the best bodyguard available for the plants in our study site.

Effect of induction on patterns of ant attendance and efficiency of anti-herbivore defence

All plants of *C. nictitans* were attended by ants at least once during our field experiment. In total, we observed ants attending our plants in 70.5% of our censuses (N = 141 of 200 censuses in 20 plants). In addition to ants, we also observed flies (N = 104), spiders (N = 8), beetles (N = 6) and wasps (N = 5) interacting with the EFNs of our focal plants. In 25.5% of our observations (36 censuses), plants were attended by more than one ant morphospecies simultaneously. A total of six ant species were observed on the plant censuses: *Brachymyrmex* sp., *Camponotus* sp., *Linepithema* sp., *Nylanderia* sp., *Pseudomyrmex* sp. and *Trachymyrmex* sp. The most frequent ants observed on the plants were *Brachymyrmex* sp. and *Nylanderia* sp., with 53.5% and 26% of all censuses, respectively (Table 3). For our analysis, we did not consider the visits by *Pseudomyrmex* sp. and *Trachymyrmex* sp. because workers of both morphospecies occurred on only one plant. In addition, we did not see any workers of *Trachymyrmex* sp. exploring or feeding on EFNs.

In general, the defence induction in *C. nictitans* did not increase any metric of ant attendance on the plants. The probability of a plant being attended by ants was similar between treatments [number of observations

Table 1. Results of the comparisons of generalized linear mixed models testing different response variables

Response variables	Model	χ^2	d.f.	P-value
Nectar volume	Saturated model × null model	12.798	3	0.005
	Saturated model × treatment + moment model	9.428	1	0.002
Sugar concentration	Saturated model × null model	4.701	3	0.19
Sugar mass	Saturated model × null model	9.692	3	0.0214
	Saturated model × treatment + moment model	7.987	1	0.005
Ant occurrence	Treatment model × null model	1.307	1	0.25
Number of ants	Treatment model × null model	0.003	1	0.85
	Saturated model × null model	274.65	7	> 0.001
	Saturated model × treatment + morphospecies model	0.685	3	0.88
	Treatment + morphospecies model × morphospecies	0.054	1	0.81
	Morphospecies model × null model	273.91	3	> 0.001
Occurrence of *Nylanderia* sp. ants	Treatment model × null model	0.3719	1	0.54
Occurrence of mimic encounters by ants	Treatment model × null model	2.5122	1	0.113
Time until encounter of mimics	Treatment model × null model	0.5963	1	0.44
Occurrence of mimic removals by ants	Treatment model × null model	0.0179	1	0.8936
Time until removal of mimics	Treatment model × null model	0.0665	1	0.7965

In the model column, we detail the predictor variables included in the two models compared. The null model did not contain any predictor variables. The model also had the interaction term between variables.

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Table 2. Diversity of ants exploring the mixed soil baits (canned sardine and honey) in our study site (Campus of Universidade Federal do ABC, São Bernardo do Campo, São Paulo)

Morphospecies	Frequency on baits (%)	Co-occurrence with other ant morphospecies	Number of aggressive behaviours	Aggressiveness index
Pheidole	0.7	1	0	0
Dorymyrmex	5.4	8	0	0
*Linepithema**	18.1	26	2	0.08
Wasmannia	6.0	9	1	0.11
*Camponotus**	4.0	6	2	0.33
*Brachymyrmex**	40.3	28	18	0.64
Solenopsis	21.5	26	18	0.69
*Nylanderia**	22.8	28	23	0.82

We calculated the ant frequency as the percentage of observations in which at least one worker from the morphospecies was present on the baits. The ant aggressiveness index was calculated as the ratio between the number of times that workers of each morphospecies displayed aggressive behaviours (shoving, biting, formic acid expulsion and stinging) and the number of observations when they shared the baits with other morphospecies.
*Ants interacting with the *Chamaecrista nictitans* (Fabaceae) plants in our study site.

(N_{obs}) = 200; treatment × null model: χ^2 = 1.31; d.f. = 1; P = 0.25): being 0.76 ± 0.43 (JA treatment) and 0.65 ± 0.48 (control plants) (Fig. 2A). Similar to the probability of ant attendance, the number of workers of any ant morphospecies on the plants was similar for both treatments (N_{obs} = 141; treatment model × null model: χ^2 = 0.03; d.f. = 1; P = 0.85; Table 1): 6.28 ± 5.92 (JA treatment) and 6.48 ± 6.34 (control plants) (Supporting Information, Fig. S4).

Our plants were attended by *Camponotus* sp. and *Linepithema* sp. workers in < 5% of our censuses. The number of ants attending plants differed among morphospecies (N_{obs} = 800; morphospecies model × null model: χ^2 = 273.91; d.f. = 3; P < 0.001; Table 1; Supporting Information, Fig. S5), with *Brachymyrmex* sp. ants recruiting more workers to the plants than the other morphospecies (7.1 ± 5.8), followed by *Nylanderia* sp. (1.8 ± 1.3). *Camponotus* sp. foraged on the plants with 1.1 ± 0.3 ants per visit, the least numerous morphospecies, followed by *Linepithema* sp. (1.5 ± 0.7 ants). However, the defence induction did not affect the number of ant visitors of the different morphospecies (N_{obs} = 800; saturated model × treatment + morphospecies model: χ^2 = 0.68; d.f. = 3; P = 0.87). Considering only the censuses in which we observed ants attending the EFNs, the probability of the most aggressive ant morphospecies, *Nylanderia* sp., feeding on EFNs was similar between treatments (N_{obs} = 141; χ^2 = 0.37; d.f. = 1; P = 0.54; Table 1; Fig. 2B): 0.38 ± 0.4 (JA treatment) and 0.35 ± 0.48 (control plants). Furthermore, the herbivory simulation did not influence the composition of ant species interacting with the plants (R^2 = –0.42, P = 0.69; Supporting Information, Fig. S6).

Similar to the patterns of ant attendance, the induction and herbivore simulation did not improve the quality of anti-herbivore defence received by the plants. Three ant morphospecies (*Brachymyrmex* sp., *Camponotus* sp. and *Nylanderia* sp.) encountered 37 larvae (37% of the larvae offered). From those, 20 were removed from the plant (54% of the larvae encountered). *Brachymyrmex* sp. encountered most larvae (N = 20). However, they removed the fewest larvae from the plant (n = 6). In turn, *Nylanderia* sp. encountered fewer larvae than *Brachymyrmex* sp. (N = 15), but they were the most efficient morphospecies at removing the larvae from the plants (N = 13; 86.7% of the larvae encountered by *Nylanderia* sp.; Table 3).

The probability of larval encounter was similar between treatments (N_{obs} = 100; treatment × null model: χ^2 = 2.51; d.f. = 1; P = 0.11; Table 1; Fig. 2C), being, on average, 0.44 ± 0.5 for plants undergoing the JA treatment and 0.3 ± 0.46 for control plants. Also, the induction had no influence on the time elapsed until ants encountered the larvae on the plants (N_{obs} = 37; treatment × null model: χ^2 = 0.60; d.f. = 1; P = 0.44; Table 1; Supporting Information, Fig. S7), which was 193.4 ± 147.9 s in plants undergoing JA treatment and 237.6 ± 151.9 s in control plants. The probability of removal of larvae from the plants was similar between treatments (N_{obs} = 37; treatment × null model: χ^2 = 0.02; d.f. = 1; P = 0.89; Table 1; Fig. 2D), being 0.54 ± 0.51 in plants undergoing JA treatment and 0.53 ± 0.52 in control plants. Likewise, induction had no effect on the time elapsed until removal of the larva by ants (N_{obs} = 20; treatment × null model: χ^2 = 0.07; d.f. = 1; P = 0.80; Table 1; Supporting Information, Fig. S8), which was 114 ± 169 s in plants undergoing JA treatment and 106 ± 205 s in control plants.

DISCUSSION

In this study, we evaluated the hypothesis that herbivory increases the attractiveness of plants bearing EFNs to high-quality ant bodyguards that,

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Table 3. Frequency of *Chamaecrista nictitans* (Fabaceae) attendance by ants and of herbivore encounter and removal by ants

Morphospecies	Frequency of ant attendance (%)		Frequency of mimic encounters (%)		Frequency of mimic removals (%)	
	CT	JA	CT	JA	CT	JA
Dolichoderinae						
Linepithema sp.	6.0	10.0	0.0	14.3	0.0	0.0
Formicinae						
Brachymyrmex sp.	50.0	57.0	20.6	48.2	12.5	15.4
Camponotus sp.	2.0	8.0	0.0	66.7	0,0	50.0
Nylanderia sp.	23.0	29.0	17.6	33.3	77.8	90.0

We used larvae of the beetle *Ulomoides dermestoides* (Tenebrionidae) as herbivore mimics on the plants. We calculated ant frequencies as the proportion of observations in which workers of a given morphospecies were observed on the plants. We calculated the frequency of mimic encounters as the proportion of the observations when ants found the herbivore mimics on the plants, considering only observations in which ant workers of the morphospecies were on the plant. Finally, we calculated the frequency of removal as the proportion of observations when ants removed the herbivore mimics from the plant, considering only observations in which ants of the morphospecies found the mimic on the plant.
Abbreviations: CT, control plants; JA, plants undergoing simulated herbivory by exogenous application of jasmonic acid (0.1 mmol/L).

in turn, increases the efficiency of plant biotic defence against herbivores. Although *C. nictitans* plants increased extrafloral nectar production after experiencing simulated herbivory, the plant-induced response did not affect any metric of ant attendance, including visitation by high-quality ant bodyguards (*Nylanderia* sp. ants). Accordingly, the simulation of herbivory also did not affect the efficiency of removal of potential herbivores by ants on the plants. In summary, we validated our assumption of the inducible nature of EFNs in plants of *C. nictitans*. However, we found no evidence that this induced response can translate into an increased probability of attendance by high-quality bodyguards and, consequently, an increased biotic defence for plants bearing EFNs. Below, we discuss the generality of the effects of induced EFN responses on patterns of ant attendance and plant biotic defence, considering factors that could modify these patterns in populations in different ecological contexts.

Two non-exclusive mechanisms can explain the null effect of defence induction on patterns of ant attendance and quality of plant defence in *C. nictitans*, namely the disproportional effect of herbivory on different extrafloral nectar traits and/or the ecological context in which our focal plants were embedded. Regarding the first mechanism, although the volume and sugar mass of the extrafloral nectar of *C. nictitans* were almost twofold higher in plants undergoing simulated herbivory, we observed no changes in nectar concentration. Considering that extrafloral nectar traits mediate the interaction between EFN-bearing plants and ants (e.g. Ness, 2003; Fagundes *et al.*, 2017; Flores-Flores *et al.*, 2018; Pacelhe *et al.*, 2019), independent variations in volume, concentration and total sugar mass of extrafloral nectar might have different meanings for plants and ants involved in the interaction. Changes in the sugar content of the extrafloral nectar secreted by plants damaged by herbivores indicate changes in the investment by the plants in this biotic defence (Gehring *et al.*, 2004; Dutton *et al.*, 2016). Therefore, from the perspective of the plant, herbivory induction can trigger an increase in energetic investment by the plant on nectar, leading to the provision of more nectar at constant quality when being attacked.

These changes in the volume and sugar concentration of the extrafloral nectar can be perceived by ants. Some studies have reported that increases in nectar volume and sugar content can lead to a higher relative value of the extrafloral nectar to ants and, consequently, to an increase in ant foraging on the plants (e.g. Hernandez-Cumplido *et al.*, 2016; Yamawo & Suzuki, 2018). However, we observed no changes in the patterns of ant attendance on plants in which defences were induced. Also, we observed that defence induction led to an increase in only one of these traits (nectar volume). This might indicate that other nectar traits exert greater influence on ant responses than the volume of nectar. From the perspective of the ant, the more concentrated the nectar, the higher the energy intake by the ant per volume of nectar ingested and, consequently, the higher the relative value of this extrafloral nectar to the attending ants (Blüthgen & Fiedler, 2004a, b; Katayama *et al.*, 2013). This might be especially true for ants occurring in habitats in which carbohydrate-based resources are scarce, such as our study site, because the costs of obtaining carbohydrates are relatively higher. If this is true, it is possible that herbivory might be more prone to generating meaningful changes in the patterns of ant attendance when increasing the concentration of the extrafloral nectar.

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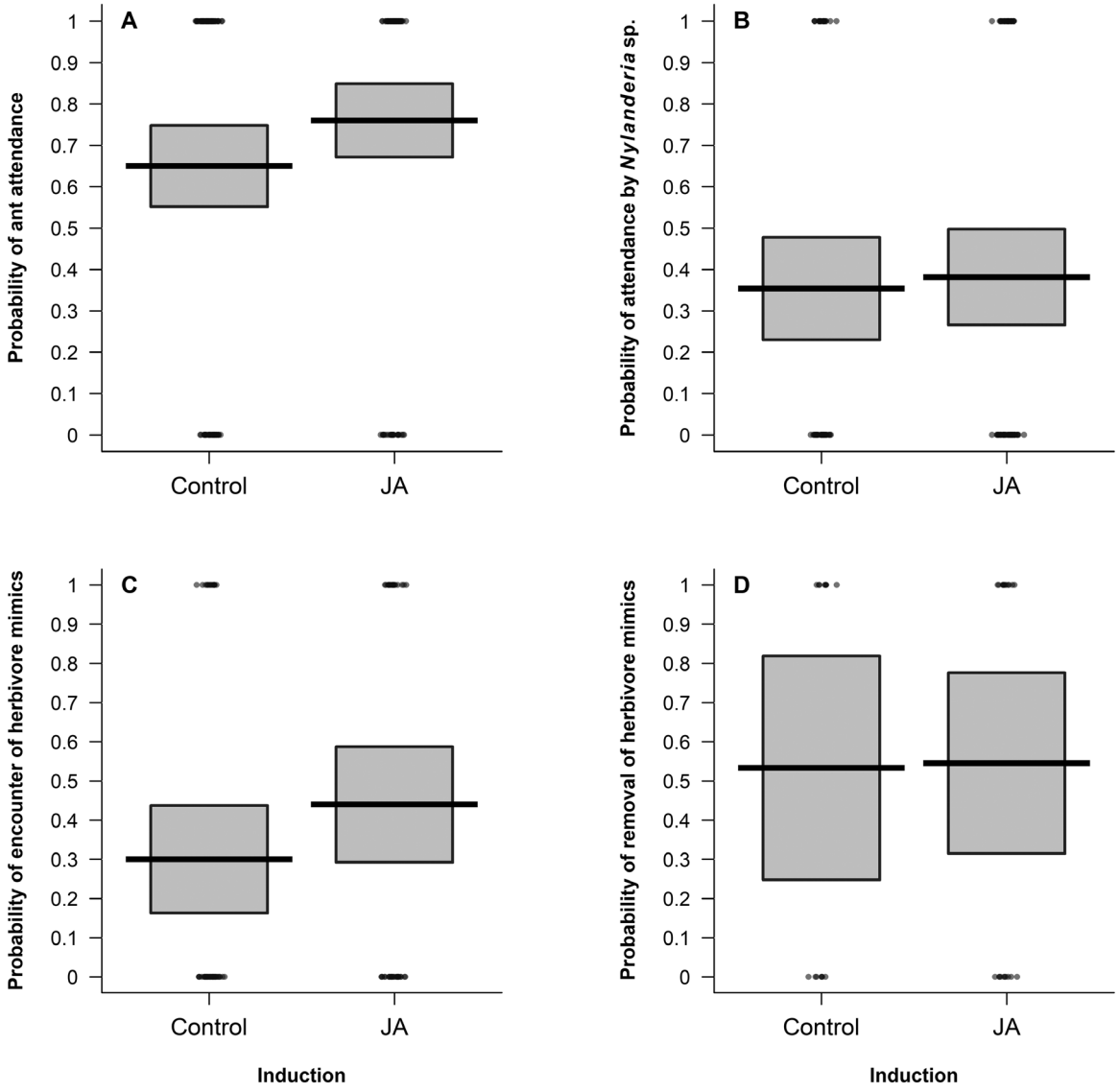


Figure 2. Influence of herbivory on patterns of ant attendance to *Chamaecrista nictitans* (Fabaceae) plants. A, probability of plants being attended by ants of any ant morphospecies [0 = plant was not attended by any ant; 1 = plant was attended by at least one ant; number of observations (N_{obs}) = 200; treatment model × null model; χ^2 = 1.31; d.f. = 1; P = 0.25]. B, probability of plants being attended by the most aggressive ants available in the habitat (*Nylanderia* sp.; 0 = plants were attended by at least one ant worker, but not by *Nylanderia* sp.; 1 = plants were attended by at least one worker from the *Nylanderia* sp. morphospecies; N_{obs} = 141; treatment model × null model: χ^2 = 0.37; d.f. = 1; P = 0.54). C, probability of ants encountering herbivore mimics (0 = mimics were not encountered by ants; 1 = mimics were encountered by ants; N_{obs} = 100; treatment model × null model: χ^2 = 2.51; d.f. = 1; P = 0.1). As herbivore mimics we used larvae of *Ulomoides dermestoides*, Tenebrionidae. D, probability of the encountered mimics being removed by ants (0 = mimics were encountered but not removed by ants; 1 = mimics were encountered and removed by ants; N_{obs} = 37; treatment model × null model: χ^2 = 0.02; d.f. = 1; P = 0.89). The plants allocated to JA treatment underwent simulation of herbivory by exogenous application of jasmonic acid, whereas the control plants did not undergo simulation of herbivory. The horizontal black line represents the mean, and the grey bars represent the 95% confidence interval. Raw data are presented as circles.

To compare and interpret the results of our research, we searched the literature for other studies evaluating the effects of herbivory induction on nectar production and the interaction with ants (Table 4). Regarding the mechanism described above, we found other studies that reported herbivory-induced effects of extrafloral nectar production on ant attendance with a similar (Ness, 2003; Kost & Heil, 2006; Jones & Koptur, 2015)

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Table 4. Studies investigating effects of herbivory on extrafloral nectar production and ant interaction

Study	Plant species	Treatment	Effect of herbivory on:			
			Volume (size effect)	Sugar concentration (size effect)	Sugar mass (size effect)	Ant interaction (size effect)
Present study	*Chamaecrista nictitans*	50% of leaf damage in 10% of leaves per plant	Positive (91%)	No effect	Positive (105%)*	No effect
Calixto *et al.* (2020)	*Qualea multifora*	10% of leaf damage	No effect	NA	No effect†	Ant abundance (248%)
		40% of leaf damage	Positive (187%)	NA	Positive (568%)†	Ant abundance (298%)
Heil (2004)	*Phaseollus lunatus*	JA application	NA	NA	Positive (1900% on leaves)‡	Herbivory reduction (NA)
Heil *et al.* (2001)	*Macaranga tanarius*	JA application	NA	NA	Positive (550%)¶	Herbivory damage (93% decrease)
		Artificial damage	NA	NA	Positive (570%)¶	Herbivory damage (93% decrease)
Hernandez-Cumplido *et al.* (2016)	*Phaseollus lunatus*	JA application	NA	Positive (220%)	NA	Ant abundance (200%)
Jones & Koptur (2015)	*Senna mexicana*	50% of leaf damage (young leaves)	Positive (NA)	Positive (36%)	Positive (129%)*	Ant abundance (197%)
Kautz *et al.* (2017)	*Prunus laurocerasus*	Artificial damage (two holes per leaf)	NA	NA	No effect‖	No effect
		Artificial damage (four holes per leaf)	NA	NA	Positive (178%)‖	No effect
		Artificial damage (six holes per leaf)	NA	NA	Positive (438%)‖	Ant abundance (4.55×); herbivore abundance (2.01× decrease)
		Artificial damage (eight holes per leaf)	NA	NA	Positive (617%)‖	Ant abundance (5.43×); herbivore abundance (2.44× decrease)
		Artificial damage (ten holes per leaf)	NA	NA	Positive (724%)‖	Ant abundance (7.29×); herbivore abundance (4.36× decrease)
Kost & Heil (2006)	*Phaseollus lunatus*	Volatile organic compounds	NA	NA	Positive (65%)‖	Ant abundance (200%); herbivory rate (224% decrease)
Meng *et al.* (2012)	*Leea glabra*	50% of leaf damage on young leaves	No effect	No effect	NA	Ant abundance (140%)
		50% of leaf damage on old leaves	No effect	No effect	NA	No effect
Ness (2003)	*Catalpa bignonioides*	Leaf damage by caterpillars	NA	NA	Positive (720%)*	Ant abundance (500–900%)

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Table 4. Continued

Study	Plant species	Treatment	Effect of herbivory on:			
			Volume (size effect)	Sugar concentration (size effect)	Sugar mass (size effect)	Ant interaction (size effect)
Ness *et al.* (2013)	*Fallopia japonica*	One-third of leaf damage	NA	NA	Positive (900%)*	Native ant abundance (180%); introduced ant abundance (700%)
Raupp *et al.* (2020)	*Qualea parviflora*	50% of leaf damage	No effect	No effect	NA	Ant abundance (44%)
		Sucking damage	No effect	Positive (~100%)	NA	Herbivore survival (25% decrease)
Yamawo & Suzuki (2018)	*Mallotus japonicus*	50% of leaf damage (peak effect between groups)	Positive (520%)	NA	NA	Ant abundance (370%)

We searched for papers in the Web of Science database (http://www.webofknowledge.com/) using all available years up to November 2020. Our search used key terms: 'extrafloral necta*' AND 'herbivor*' AND 'induc*' AND 'ant'. The search resulted in 89 papers. We read the abstracts and included in the table only papers in which the effect of herbivory induction was evaluated on extrafloral nectar production and ant visitation or defence. Treatment is the type of herbivory simulation or use of true herbivores in each study. We considered herbivory effects on sugar mass papers that presented: *total sugar mass (in milligrams); †calories (in kilocalories) calculated based on the sugar content; and total sugar standardized by leaf mass (‡dry weight or ⁋fresh weight) or ¶leaf area. Size effects were calculated as the fold increase or decrease based on mean differences between groups when available in the paper.

or even lower (Heil *et al.*, 2001, 2004; Ness *et al.*, 2013; Kautz *et al.*, 2017) magnitude of the effect of herbivory on total sugar content than we observed here. However, most of these studies used only the total sugar mass as a descriptor of the extrafloral nectar response to herbivory (Table 4). Therefore, it is not possible to conclude whether the ant attractiveness and/or defence reported in these studies are caused by changes in total sugar content or other unexplored extrafloral nectar traits, such as nectar volume or sugar concentration. Extrafloral nectar volume and sugar concentration tend not to be collected or reported because they are more susceptible to microclimatic variations than estimations of total sugar mass (Heil *et al.*, 2001). Here, we collected the extrafloral nectar in controlled greenhouse conditions, and our measurements of nectar volume and concentration were not correlated with temperature. These aspects minimize our chances of biased collection or interpretation of patterns involving these nectar traits. Therefore, despite the lack of a more detailed description of changes in extrafloral nectar in response to herbivory in the literature, our rationale is that the role of herbivory-induced changes in the volume and concentration of extrafloral nectar on ant attractiveness should not be overlooked but investigated explicitly using a more integrative approach.

Regarding the measures of ant attendance on plants, it is also essential to consider that the herbivory-induced studies focus on different moments and benefits of the interaction with ants. Most studies measure the abundance of ants on plants (Table 4), whereas others also evaluate the intensity of herbivory, rates of herbivory and abundance of herbivores (Table 4). These differences can also make further comparisons among studies difficult. Therefore, the comparative investigation of each outcome descriptor of ant–plant–herbivore interactions is extremely important to understand the impact of inducible EFN defence on the full range of benefits that plants receive in this mutualism.

A second mechanism that can explain our findings is that changes in patterns of plant attendance by ants in response to herbivory can be context dependent. We performed our study in a human-disturbed area where the availability of alternative carbohydrate sources to ants (floral nectar, honeydew, fruit-pulps or seed arils) is relatively low (H. D. Chinarelli, personal observation). Likewise, the local ant richness is typically lower in anthropized areas than in preserved areas (e.g. Yamaguchi, 2004; Leal *et al.*, 2012). In such conditions, ant species (especially the dominant ones) that manage to establish in these areas could intensify their foraging on the few sources of carbohydrates available because the relative value of any carbohydrate source to ants would already be high in this habitat. For example, *Nylanderia* sp., the most aggressive bodyguard in our study site, occupied

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one-quarter of the baits that we used to evaluate the aggressiveness of the local ant community (Table 2). Likewise, this ant morphospecies attended roughly one-quarter of JA and control plants (23% on control plants and 29% and on JA-treated plants). If ant species in this habitat are intensively foraging the few plant resources available, the increases in the relative quality of the extrafloral nectar secreted by the plants after the herbivory simulation would not increase the probability of ant attendance on these plants.

Although our study was conducted in a human-disturbed area, it is essential to note that it describes the response of ants and plants bearing EFNs to herbivory in a habitat where both mutualists occur naturally. Disturbed sites are colonized by ruderal species bearing EFNs that naturally spread and become abundant in these areas (winner species; Leal *et al.*, 2015). In a similar manner, some of the most classical studies on the effects of ant attendance on the performance of plants bearing EFNs were carried out in the Brazilian Cerrado (e.g. Oliveira *et al.*, 1987; Del-Claro *et al.*, 1996; Oliveira & Pie, 1998), a vegetational mosaic of savanna areas frequently affected by natural and anthropic fires and other human activities (Coelho *et al.*, 2020; Lopes dos Santos *et al.*, 2021). More specifically, populations of *C. nictitans* occur naturally in our study site, meaning that our results describe what should be happening in this population in terms of mutualism functioning. Therefore, the absence of an ant response to herbivory-induced defences in *C. nictitans* described here must not be interpreted as a local or artificial result. Instead, these results highlight that the influence of induced responses to herbivory on patterns of attendance by ants on plants bearing EFNs is conditional. The absence of this influence might be more common than expected in carbohydrate-poor disturbed habitats, in which most EFN-bearing plants occur naturally.

In summary, our results showed that changes in secretion of extrafloral nectar after herbivory did not influence the attractiveness of *C. nictitans* plants to high-quality ant bodyguards or ants in general. Consequently, *C. nictitans* plants do not benefit more from ant anti-herbivore defence when under herbivory attack. We argue that increasing biotic protection by high-quality ant bodyguards after herbivory is more likely in plants in which herbivory induces changes in extrafloral nectar volume and sugar concentration, thus improving the quantity and relative quality of the nectar to ants, especially high-quality partners. This pattern might also be true for other examples of differential investment in extrafloral nectar production, such as that throughout leaf development stages (see Calixto *et al.*, 2015). The foraging response of ants to differential production of nectar in plants in particular conditions or at particular phenological stages might also be dependent on increases in extrafloral nectar volume and sugar concentration. We also argue that the response of ants to induced plant defence is context dependent, and in conditions of low availability of carbohydrate resources, the herbivory-induced response might not trigger significant changes in ant attendance because ant foraging on these resources would already be at a maximum.

However, the fact that there are spatial and temporal situations in which herbivory-induced defences do not improve plant defence raises an important question: why are herbivory-induced defences maintained in EFN-bearing species occurring in carbohydrate-poor habitats? Considering that herbivory-induced changes in extrafloral nectar occur in several non-phylogenetically related plant species (see Heil, 2015; Table 4), it is possible that this strategy of defence can provide other benefits to EFN-bearing plants not investigated here. For example, a herbivory-induced response of plants in the volume of extrafloral nectar, like in *C. nictitans*, might increase the constancy of ant attendance and defence over time. Also, carbohydrate-poor habitats are not permanently poor. On the contrary, ecological succession can improve the local diversity of resources and changes in ant assemblages over time. Therefore, other types of benefits related to quantitative changes in nectar secretion and temporal changes of the context of the interaction need to be investigated further, in order to disentangle the effects of the different descriptors of extrafloral nectar on the attractiveness to ant bodyguards and, consequently, on the quality of anti-herbivore defence received by the plants.

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design, data analyses, supervision and writing. L.C.L. contributed to conceptualization, funding acquisition, experimental design, data analyses, supervision and writing. The authors have no conflicts of interest to declare.

DATA AVAILABILITY

The data are available via the link: https://github.com/lacaleal/Chinarelli-et-al.-2021---BJLS.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1. Pearson correlation coefficient estimated between extrafloral nectar descriptors and the temperature at the time of collection of the extrafloral nectar.

Table S2. Statistical effects of each predictor variable included in the models evaluating the influence of herbivory simulation on the volume [in microlitres per active extrafloral nectary (EFN)], sugar content (in milligrams per active EFN) and sugar concentration (in degrees Brix) of extrafloral nectar secreted by *Chamaecrista nictitans* (Fabaceae). The statistics were calculated by fixing one variable and analysing the interaction of the other variable with each level of the fixed variable with the testInteractions function of the package PHIA (Rosario-Martinez, 2015). In the first sampling, both plant groups had the extrafloral nectar sampled before simulation of herbivory. In the second sampling, we simulated herbivory in half the plants with the exogenous application of jasmonic acid (JA treatment), while the other plants were treated as the control group.

Table S3. Random effects estimated in the saturated generalized linear mixed models (GLMMs) applied to the different response variables. To select the GLMMs, we compared a saturated model with a null model. When the models differed, we removed the interaction and the predictor variables and compared the models with the saturated model and the null model. We repeated this until the removal of one predictor variable (or the interaction between variables) from the saturated model affected the comparison with the saturated model.

Figure S1. Absence of relationship between the mean volume of extrafloral nectar per active extrafloral nectary (EFN) (in microlitres) secreted by *Chamaecrista nictitans* (Fabaceae) plants and temperature (in degrees Celsius).

Figure S2. Absence of relationship between the probability of *Chamaecrista nictitans* (Fabaceae) plants bearing extrafloral nectaries (EFNs) being attended by ants (0 = plant was not attended by any ants; 1 = at least one ant attended the plant) and temperature ($z = -0.887$, $P = 0.37$).

Figure S3. Absence of relationship between the probability of *Chamaecrista nictitans* (Fabaceae) plants bearing extrafloral nectaries (EFNs) being attended simultaneously by ants of different morphospecies (0 = plant was

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attended by one ant morphospecies; 1 = plant was attended by two or more ant morphospecies simultaneously) and temperature ($z = -0.757$, $P = 0.45$).

Figure S4. Absence of relationship between the number of ants visiting *Chamaecrista nictitans* (Fabaceae) plants bearing extrafloral nectaries (EFNs) and herbivory ($\chi^2 = 0.03$, d.f. = 1, $P = 0.85$). Plants undergoing the jasmonic acid (JA) treatment received exogenous application of jasmonic acid, representing our simulation of herbivory, whereas control plants were not manipulated. The horizontal lines represent the means, and the grey bars represent the 95% confidence interval. Raw data are presented as circles.

Figure S5. Relationship between the number of ants visiting *Chamaecrista nictitans* (Fabaceae) plants bearing extrafloral nectaries (EFNs) and the morphospecies of ant visitors ($\chi^2 = 273.91$, d.f. = 3, $P < 0.001$). Plants undergoing the jasmonic acid (JA) treatment received the exogenous application of jasmonic acid, representing our simulation of herbivory, whereas control plants were not manipulated. The horizontal lines represent the means, and the grey bars represent the 95% confidence interval. Raw data are presented as circles.

Figure S6. Composition of ant species visiting *Chamaecrista nictitans* (Fabaceae) plants bearing extrafloral nectaries (EFNs) in the herbivory simulation group and the control group. Plants undergoing the herbivory simulation (JA) received exogenous application of jasmonic acid, whereas the control plants were not manipulated. Open circles and filled circles represent ant visitors sampled on JA plants and control plants, respectively. The figure produced through non-metric multidimensional scaling (NMDS) ordination using the Bray–Curtis dissimilarity matrix of the frequency of different ant species on the baits (stress = 0.08, $R^2 = -0.42$, $P = 0.69$).

Figure S7. Absence of relationship between the time elapsed (in seconds) until ants found the mimics of herbivores (larvae of *Ulomoides dermestoides*, Tenebrionidae) and the simulation of herbivory ($\chi^2 = 0.60$; d.f. = 1; $P = 0.44$). Plants undergoing jasmonic acid (JA) treatment received exogenous application of jasmonic acid, representing our simulation of herbivory, whereas the control plants were not manipulated. The horizontal lines represent the means, and the grey bars represent the 95% confidence interval. Raw data are presented as circles.

Figure S8. Absence of relationship between the time elapsed (in seconds) until ants removed the mimics of herbivores (larvae of *Ulomoides dermestoides*, Tenebrionidae) and the simulation of herbivory ($\chi^2 = 0.07$, d.f. = 1, $P = 0.8$). Plants undergoing the jasmonic acid (JA) treatment received exogenous application of jasmonic acid, representing our simulation of herbivory, whereas the control plants were not manipulated. The horizontal lines represent the means, and the grey bars represent the 95% confidence interval. Raw data are presented as circles.

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